

# Deep Pattern Builder

Indicators & Analysis / Deep Indicators | advanced | 7 min read | DeepCharts Help Center | July 16, 2026

Deep Pattern Builder is the most advanced indicator in the DeepCharts suite. It lets you construct custom trading logic out of relational conditions — comparisons like  $A + B > C + D$  over candle values, volume, delta, POC references or other indicators' outputs — and then run automated backtests on that logic with the built-in simulation engine.

In practice it is two tools in one: a signal engine that plots markers or background highlights wherever your conditions are true, and a backtester that turns those signals into simulated entries with stops, targets and trailing logic.

## What it is

Each condition in the builder takes up to four inputs (**A, B, C, D**), joins A with B and C with D using math operators, and compares the AB block against the CD block with one logical operator. Conditions can be combined with AND/OR logic, filtered by an integrated imbalance filter, visualized on the chart, wired to alerts, and fed into a simulated entry/exit model for backtesting.

[ SCREENSHOT PLACEHOLDER ] Chart with Deep Pattern Builder applied — signal markers plotted on candles where the custom conditions fired, with a couple of background-highlighted bars visible  
deep-pattern-builder-chart-signals.png

## When to use it

- You have a repeatable idea ("close above the POC with positive delta") and want the platform to mark it on every bar, live and historically.
- You want to reference other indicators' values (for example a moving average) inside your conditions.
- You want statistical evidence: the built-in backtest simulates entries, stops and targets over historical data.
- You want alerting on multi-part logic that no single stock indicator covers.

## Quick start

1. Apply the **Deep Trades** indicator to the chart first — the builder workflow depends on it (see Deep Trades).
2. Click the **Indicator** icon, search for **Deep Pattern Builder** and click **+** to add it.
3. Click the **Settings** icon next to Deep Pattern Builder.
4. Build a first condition. Example — "delta-positive close above the open": **Condition 1**: Input A = **Reference** → **Close**, Logical Operator **>**, Input C = **Reference** → **Open**. **Condition 2**: Input A = **Reference** → **Delta Volume**, Logical Operator **>**, Input C = **Constant** → **0**. Combine with **AND (Simple)**.
5. Enable **Calculate on Close** so signals evaluate only when the candle completes, without intrabar flicker.
6. Under **Plot Settings**, choose a **Plot Type** (start with **Marker**) and a **Plot Price** placement.
7. Click **Apply and Save** and check that markers appear on historical bars where the logic is true.

[ SCREENSHOT PLACEHOLDER ] Deep Pattern Builder settings dialog showing one enabled condition with Input A set to Reference Close, the Logical Operator dropdown, Input C set to Reference Open, and the Combine Mode selector below  
deep-pattern-builder-condition-setup.png

## How to read it

A signal (marker or highlighted background) means every part of your combined logic evaluated true on that bar — nothing more. The tool plots your idea; judging the idea is what the backtest is for.

- **Markers** are placed at the **Plot Price** you chose (High = above the candle, Low = below, Current = at the close, Mid Price = wick center), colored per side via the subgraph colors.
- **Flickering signals during a live bar** mean the condition is oscillating intrabar — enable **Calculate on Close** if you only want confirmed signals.
- **Backtest report**: after running a backtest, the report shows how the simulated entries performed under your stop/target model. Iterate: adjust the conditions or risk settings, re-run, compare against previous saved reports.

## Settings reference

Grouped as in the settings dialog.

### Condition

| Setting                  | What it does                                                                                     |
|--------------------------|--------------------------------------------------------------------------------------------------|
| <b>Enable</b>            | Activates the condition.                                                                         |
| <b>Inputs A, B, C, D</b> | Up to four input slots per condition; B and D are optional (set unused slots to <b>Unused</b> ). |
| <b>Math Operator 1</b>   | +, −, ×, / between inputs A and B.                                                               |
| <b>Math Operator 2</b>   | +, −, ×, / between inputs C and D.                                                               |
| <b>Logical Operator</b>  | Compares the AB block with the CD block: >, >=, =, <> (not equal), <, <=.                        |

### Input types

| Input Type       | What it does                                                                                                                                                                                                                                |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Unused</b>    | Disables the input slot.                                                                                                                                                                                                                    |
| <b>Reference</b> | Candle components: Open, Close, High, Low, Bid Volume, Bid NT, Ask Volume, Ask NT, Total Volume, Total NT, Delta Volume, Delta NT, Low–High Ticks, Open–Close Ticks, POC Price %, POC Total Volume, POC on Shadow, Cumulative Delta Volume. |
| <b>Indicator</b> | Live values from another applied indicator — requires that indicator's <b>Indicator ID</b> and <b>Subgraph ID</b> (the specific value line to read).                                                                                        |
| <b>Constant</b>  | A fixed numeric value you type in.                                                                                                                                                                                                          |

| Option              | What it does                                                                                                                                                                      |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Previous Bar</b> | Pulls the input's value from the prior candle instead of the current one — enables bar-over-bar logic such as "delta rising" (A = Delta Volume > B = Delta Volume, Previous Bar). |

### Combine modes

| Mode                       | What it does                                                                            |
|----------------------------|-----------------------------------------------------------------------------------------|
| <b>OR (Simple)</b>         | Any single condition being true fires the signal.                                       |
| <b>AND (Simple)</b>        | All conditions must be true.                                                            |
| <b>AND + OR (Advanced)</b> | Nested groupings such as (Condition 1 OR Condition 2) AND (Condition 3 OR Condition 4). |
| <b>Calculate on Close</b>  | Evaluates conditions only after the candle closes — no intrabar flicker.                |

### Imbalance filter

| Setting          | What it does                                                                                                                                                   |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Imbalance filter | An integrated filter whose configuration mirrors the Imbalance Tracker indicator's settings — restricts signals to bars that also meet the imbalance criteria. |

### Plot Settings

| Setting         | What it does                                                                                                                        |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Plot Type       | <b>Marker</b> (signal marker only), <b>Background</b> (candle background highlight), or <b>Series &amp; Background</b> (both).      |
| Days to Load    | How many historical days to calculate signals for.                                                                                  |
| Plot Price      | Marker placement: <b>High</b> (above candle), <b>Low</b> (below candle), <b>Current</b> (at close), <b>Mid Price</b> (wick center). |
| Color / Opacity | Marker or background color and background transparency.                                                                             |

### Alert Settings

| Setting         | What it does                                                                                                                             |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Alert selection | Alerts must be created first via <b>Options</b> → <b>Add Alert</b> ; only then do they appear in the builder's alert list for selection. |

### Simulation (backtesting)

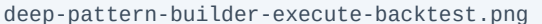
| Setting                                                   | What it does                                                                                                               |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Enable                                                    | Activates simulation mode.                                                                                                 |
| Entry Type                                                | <b>Market</b> (immediate entry), <b>Half Bar</b> , or <b>Ticks from Close</b> (limit order X ticks from the candle close). |
| Side                                                      | <b>Buy</b> or <b>Sell</b> .                                                                                                |
| Allow Multiple Entries                                    | Permits concurrent positions.                                                                                              |
| Entry Expiration (Bars)                                   | How many bars a pending entry stays valid.                                                                                 |
| Stop (Ticks)                                              | Fixed stop-loss distance.                                                                                                  |
| Trailing Stop — Enable / Trigger (Ticks) / Offset (Ticks) | Activates a trailing stop; Trigger is the profit threshold before it engages, Offset the gap it maintains behind price.    |
| Targets                                                   | Multiple profit targets in ticks.                                                                                          |
| Move to Break Even (B/E)                                  | Moves the stop to entry price when a target is reached.                                                                    |

### Subgraph

| Setting         | What it does                         |
|-----------------|--------------------------------------|
| Color / Color 2 | Bid-side and ask-side marker colors. |
| Subgraph style  | Marker style selection.              |

## Running a backtest

1. Configure conditions, plot and simulation settings, then click **Apply and Save**.
2. Right-click the chart.
3. Select **Autotrading** → **Execute Backtest**.
4. Review the report. Via right-click you can also clear results, view previous backtests and save reports.

[ SCREENSHOT PLACEHOLDER ] Right-click context menu on the chart with Autotrading expanded and Execute Backtest highlighted, with a completed backtest report visible 

## Tips and common mistakes

- **Backtest option missing on right-click?** Follow the full workflow in order: Deep Trades applied, Deep Pattern Builder added and configured, **Apply and Save** clicked — then right-click → **Autotrading** → **Execute Backtest**.
- **Alert not listed in the builder?** Alerts must exist before the builder can use them — create them first via **Options** → **Add Alert**.
- **Signals flicker intrabar?** Enable **Calculate on Close**.
- **Referencing another indicator returns nothing?** Check both the **Indicator ID** and the **Subgraph ID** — the Subgraph ID selects which value line of that indicator you read.
- **Start with one condition.** Verify each condition plots sensibly on its own before combining with AND/OR — debugging nested logic with four inputs per condition is much harder.
- **Bar-over-bar logic needs Previous Bar.** "Delta rising" is current Delta Volume > Delta Volume with **Previous Bar** enabled on the second input, not two separate conditions.

## Related articles

- Deep Trades
- Imbalance Tracker
- Orderflow 101: Bid, Ask, Aggressor and Delta
- Different Types of Input Data for Indicators
- How to Change the Layout of Indicators
- Dynamic POC